MIL-C-46240A (MU)
21 July 1965
SUPERSEDING
MIL-C-46240 (Ord)
6 July 1960

MILITARY SPECIFICATION
CARTRIDGE, THRUSTER, M67A1

1. SCOPE

1.1 This specification covers the Cartridge, Thruster, M67A1 for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

Munitions Command

D8597730	- Cartridge, Thruster, M67A1 Assembly
F8797726	- Carton, Packing, Ammunition, for Cartridge M67A1
F8797727	- Box, Packing, Ammunition, for Cartridge M67A1
IEL-8597730	- Inspection Equipment List for Cartridge, M67A1

FSC 1375

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, assemblies, chemical and physical requirements shall conform to the applicable drawings, referenced specifications and the following:

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Protective finish.- When protective finish is required by component details, finish requirements shall be as required on the drawing and referenced specification and shall apply to all surfaces of the assembled cartridge regardless of contour configuration.

3.4 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer or propellant, or derangement of components.

3.5 Functional requirements.

3.5.1 The cartridge shall not fail to fire.

3.5.2 Powder charge.- The propellant charge shall be fixed by the contractor to produce the specified ballistic requirements. (See 6.1.)

3.5.3 Ballistics.- The cartridge when tested in either the M6 or M7 Thruster, temperature conditioned at minus 65 degrees Fahrenheit (F.) to minus 70 degrees F. and at plus 160° F. to plus 165° F., shall comply with the applicable requirements in Table I:

Table I

Thruster M6	Thruster M7
Shall impart to a 60 pound weight, a minimum initial thrust of 400 pounds while propelling the 60 pound weight 1 1/2 inches plus or minus 1/16 inch on a horizontal plane. Ignition Delay: 50 milliseconds The thruster shall be fired by the gas pressure generated from an M3A1 Initiator with 20 feet of aircraft hose, size 4, MS28741.	Shall impart to an 11.6 pound weight, a minimum initial thrust of 500 pounds at the end of 1/2 inch plus or minus 1/16 inch stroke, and a minimum initial thrust of 260 pounds at the end of 2 1/2 inches plus or minus 1/16 inch stroke while propelling the 11.6 pound weight on a horizontal plane. Time to complete thruster stroke shall not exceed 70 milliseconds. Ignition Delay: 65 milliseconds The thruster shall be fired by the gas pressure generated from an M3A1 Initiator with 15 feet of aircraft hose, size 4, MS28741.

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3.6 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, specifications and the following:

3.6.1 Metal defective.- All metal parts shall be free from cracks, splits, laminations, perforations, porosity or other metal defects.

3.6.2 Burr.- No cartridge shall have a burr which might interfere with assembly or function of the item which might be injurious to personnel handling the cartridge.

3.6.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, oil, rust, corrosion or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2. Initial production sample.-Prior to the initiation of regular production, the contractor shall submit an initial production sample to a designated Government approved facility for evaluation. The sample shall consist of 35 cartridges and 10 sets of unassembled components, including explosive components. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used in regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Lot.

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4.3.1.1 Submission of product. - The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot. - Unless otherwise specified by the procuring activity, an ammunition lot shall consist of 600 cartridges, maximum, exclusive of test samples and shall consist of:

a. Components produced by a homogeneous process.
b. Propellant from not more than one lot.
c. Primers from not more than one lot.

4.3.2 Examination. - One hunderd percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance standard finish blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

Table II

Lot Size	Sample Size	Acceptance Numbers Major	Minor
1 - 300	100% Examination	0	0
301 - 1300	110	1	2
1301 - 3200	150	2	3

Table III

No. of Defects in Class	Total Major Defects Allowable
2 - 3	1
4 - 7	2
8 - 19	3
No of Defects in Class	**Total Minor Defects Allowable**
2 - 3	2
4 - 7	3
8 - 16	4

4.3.2.1 Classification of defects.

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4.3.2.1.1 Cartridge, Thruster, M67A1 Assembly, Drawing D8597730.

Defect	Method of Inspection
CRITICAL	
1. Any part missing	X-ray/Visual 1/
2. Improper assembly	X-ray/Visual 1/
3. Primer not fully seated	X-ray/Visual 1/
4. Primer pellet missing	X-ray/Visual 1/
5. Anvil missing, cocked, broken or inverted	X-ray/Visual 1/
6. Depth to primer pocket face	Gage
7. Primer seating depth	Gage
8. Mixed ammunition	Visual
9. Metal defective	Visual
10. Crimp missing or incorrect	Visual
11. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Diameter of body, max.	Gage
103. Diameter of head, max.	Gage
104. Length to datum diameter, max.	Gage
105. Primer crimp missing	Visual
106. Protective finish inadequate or defective	Visual
107. Foreign matter	Visual
108. Burr	Visual
MINOR	
201. Total length	Gage
202. Dent	Visual
203. Draw scratch	Visual
204. Fold, wrinkle, buckle or bulge	Visual
205. Varnish inadequate or missing	Visual
206. Surface finish improper	Visual

1/ The cartridges shall be x-rayed with the long axis of the cartridges perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

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4.3.2.1.2 Carton, Packing Ammunition for Cartridge M67A1, Drawing F8797726.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Contents loose	Manual/Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damaged through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.1.3 Box, Packing, Ammunition, for Cartridge, M67A1, Drawing F8797727.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Contents loose	Manual/Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.- The follwoing tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable specification listed on the drawing.

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4.3.3.2 Airtightness.- Each cartridge shall be subjected to the minimum pressure differential. Failure to meet the requirements of the drawing shall be cause for rejection of the cartridge.

4.3.3.3 Vibration test initial production sample.- Twelve of the 30 assembled x-rayed cartridges of the initial production sample, properly identified, shall be subjected to vibration as directed in 3.4. After the vibration test the twelve vibrated cartridges shall be x-rayed as directed in note 1/ of 4.3.2.1.1. Evidence of any disintergation of the primer, propellant or igniter or derangment of any component shall be cause for rejection of the lot.

4.3.3.4 Vibration test production sample.- When it becomes necessary to make a change in the lot of propellant, or igniter or primers, after production is initiated, 10 assembled cartridges from the 1st production lot thereafter shall be tested in accordance with 4.3.3.3.

4.3.3.5 Sealant application.- Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.3.6 Functional.- The following ballistic tests will be performed by the Government:

4.3.3.6.1 Initial production sample.- Thirty units shall be tested to determine compliance with the requirements of 3.5, as applicable. The units shall be temperature conditioned as specified in Table IV and tested in accordance with 4.4.4.

4.3.3.6.2 Production lot sample.- Ten cartridges from each production lot shall be tested to determine compliance with the requirements of 3.5, as applicable. The units shall be temperature conditioned as specified in Table IV and tested in accordance with 4.4.4.

TABLE IV

Temperature	Quantity	Method
- 65° to - 70°F.	15 (see 4.4.4.1)	Thrust, Stroke and Time
+160° to +165°F.	15 (see 4.4.4.1)	Thurst, Stroke and Time
- 65° to - 70°F.	5	Thrust, Stroke and Time
+160° to +165°F.	5	Thrust, Stroke and Time

Rejection criteria.- Failure of any unit to comply with the requirements, as applicable, shall be cause for rejection of the initial production sample or production lot, as applicable.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8597730.

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4.4 Test methods and procedures.

4.4.1 Protective finish.- Test procedures for protective finish shall be in accordance wit the applicable specification referenced on the drawing.

4.4.2 Airtightness.- The method for determining airtightness shall be by use of equipment listed on the inspection equipment list, except as provided in 4.3.

4.4.3 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.4 Functional.

4.4.4.1 Initial production sample.- The 30 cartridges shall include the 12 vibrated cartridges and shall be tested in 2 groups of 15 each; each group shall contain 6 vibrated cartridges. The test units containing the vibrated cartridges shall be properly identified.

4.4.4.2 Ballistics.- Cartridges shall be loaded into acceptable M6 or M7 Thrusters with the end caps to cover the gas port during the temperature conditioning period. The tests' units shall be rigidly supported so that the entire thrust is borne by the thruster mounting points. The units shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the tests' units shall be reconditioned at the original temperature.

4.4.4.3 Temperature conditioning.- A simulated thruster and cartridge with a temperature sensitive element, placed in the dummy cartridge, shall be stored in a temperature conditioning chamber at the same time as the test units. The temperature conditioning shall be a minimum of one hour longer than the time necessary for the temperature of the simulated cartridge to reach the required test temperature. The total conditioning time, including reconditioning at plus 165 degrees F., shall not exceed 18 hours.

4.4.4.4 Sealant application.- The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the joining surfaces at the junction of the cartridge head and case sidewall and that the propellant cavity is free of sealant.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings F8797726 and F8797727.

5.2 Data cards.- Data card information shall be in accordance with MIL-A-2550.

6. NOTES

6.1 Advisory loading data:

Black Powder, Grade A4	23 ± 1/2 grains

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1375-A865

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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